AIR UNIVERSITY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

EXPERIMENT NO 10

Lab Title: Flip Flops

Student Name: Muhammad waqas Reg. No: 240756

Objective: To know the ifference between combinational and sequential circuit to understand flip flops and its design and teir practical working

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____

EXPERIMENT 10

Flip Flops

Objectives:

- To know difference between combinational and sequential logic.
- To understand flip flops and its design.
- To have practical knowledge and working of:
    - Positive Edge triggered D – flip flop
    - Positive Edge triggered JK flip flop
    - Make a toggle and D flip flop using JK flip flop

Equipment required:

- ICs as required
- Digital Electronics Trainer

Background Knowledge:

Combinational Logic:

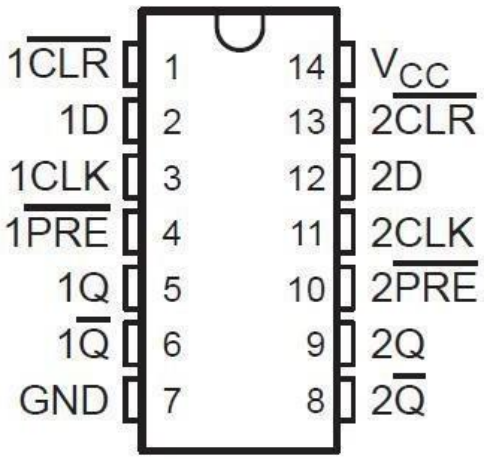
Combinational logic circuits implement Boolean functions. Boolean functions are mappings of input bits to output bits. These circuits are functions of input only.

Sequential Logic:

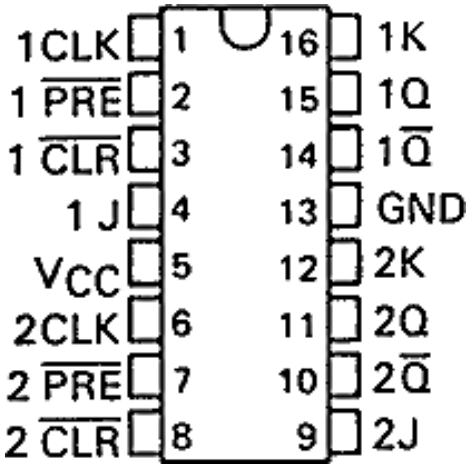
Unlike combinational logic, sequential circuits have state, which means basically, sequential circuits have memory. A sequential circuit uses flip flops. A flip-flop or latch is a circuit that has two stable states and can be used to store state information.

In this lab we will use ICs 74xx74 and 74xx76, which are D and JK flip flop respectively. Their pin configuration is given below:

74xx74



74xx76

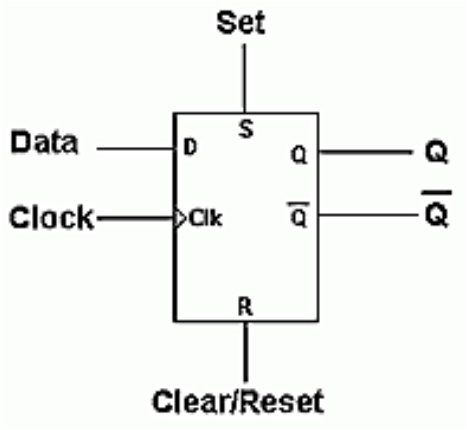


Task 1:

Use D flip flop from IC 74xx74 and complete its truth table.

Steps:

1. Use 555 timer or CLK on your digital trainer kit as clock source.
2. Observe above waveform on digital oscilloscope or digital electronics trainer.
3. Apply different inputs on the flip flop and verify the results.

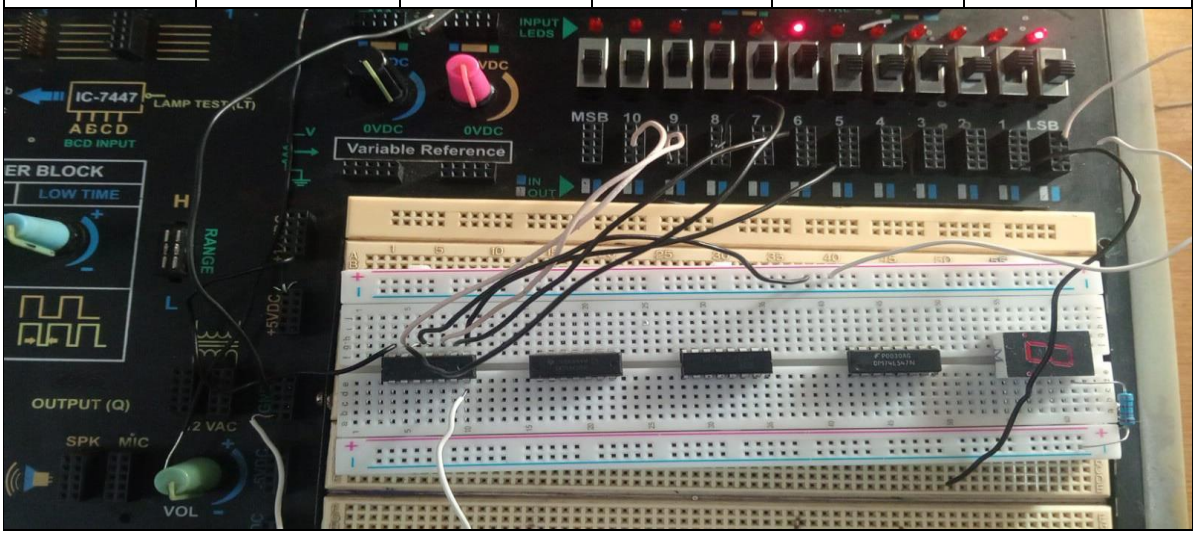


Truth Table:

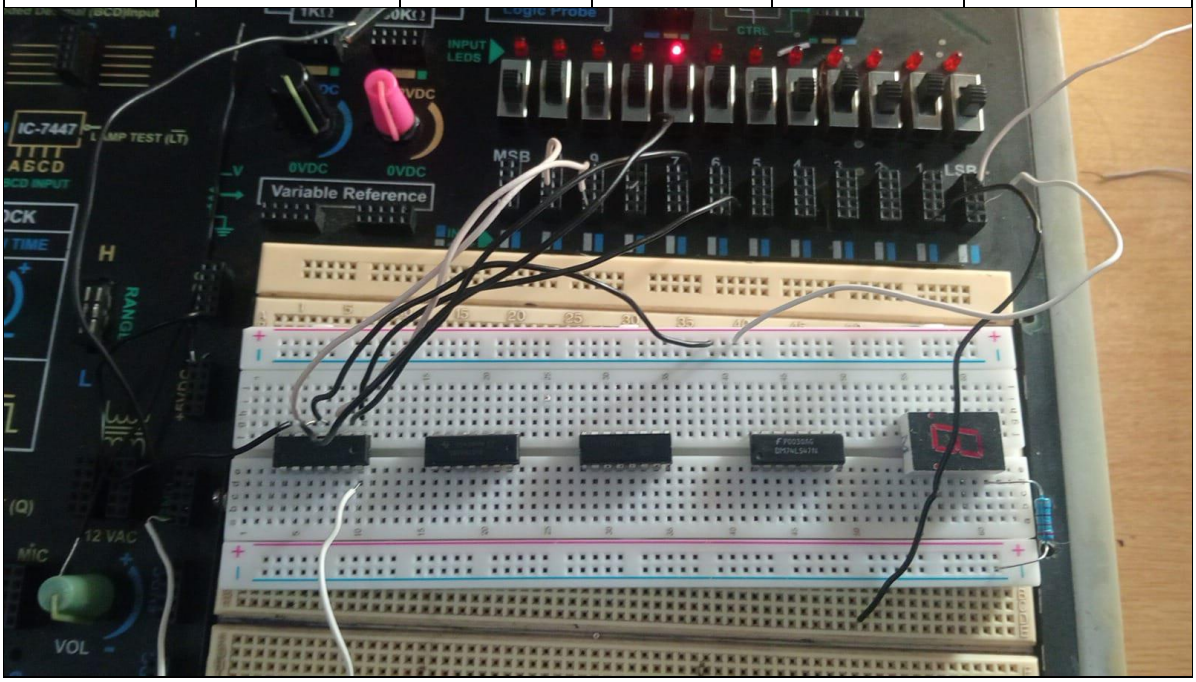
Inputs				**Outputs**	
PR	**CLR**	**CLK**	**D**	**Q**	**Q'**
0	1	X	X	1	0
1	0	X	X	0	1
0	0	X	X	X	X
1	1	↑	1	1	0
1	1	↑	0	0	1
1	1	↓	X	Qo	Qo'

Simulation Activity:

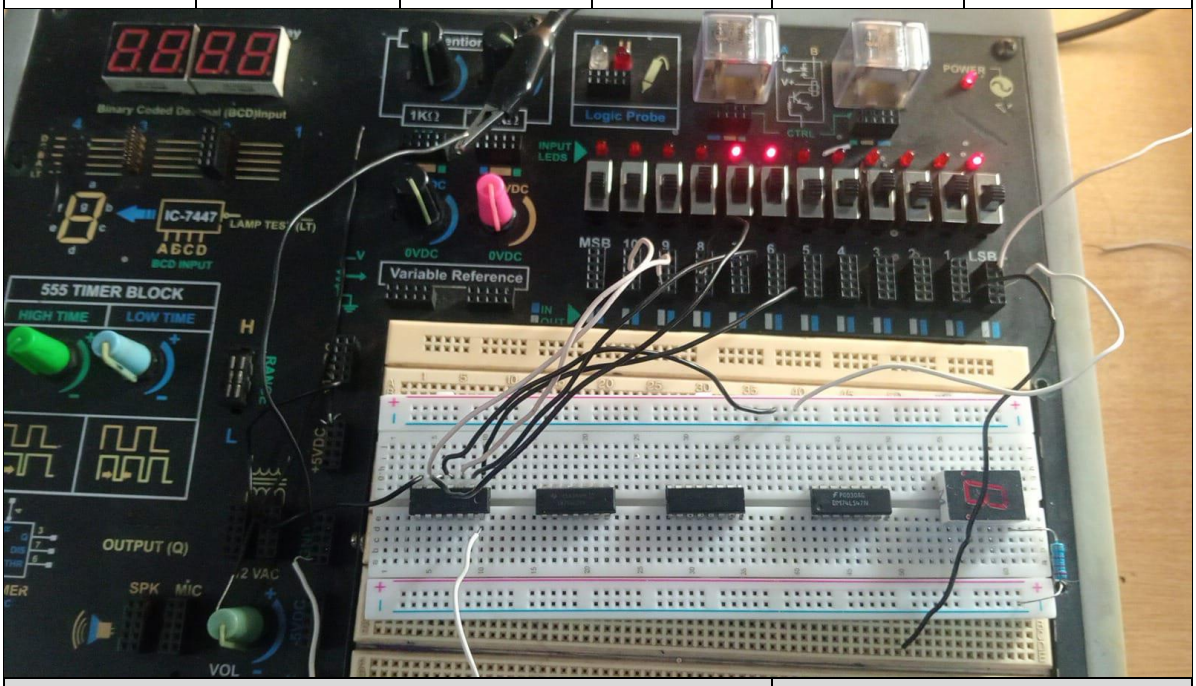
Inputs				Outputs	
PR	**CLR**	**CLK**	**D**	**Q**	**Q'**
0	1	X	X	1	0



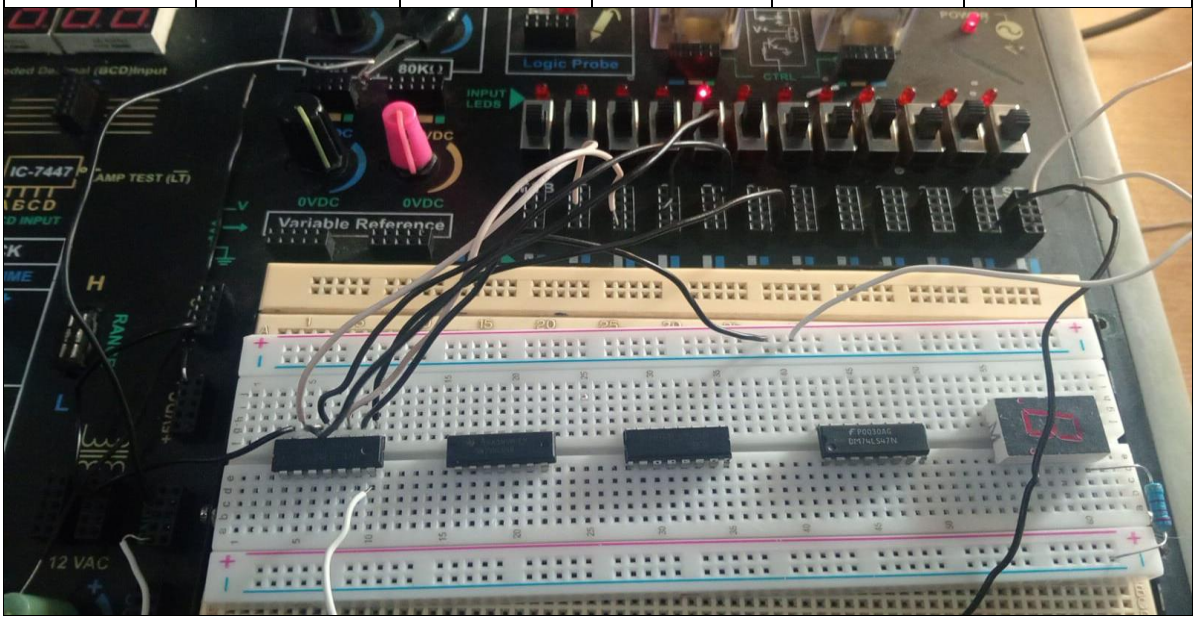
Inputs				Outputs	
PR	**CLR**	**CLK**	**D**	**Q**	**Q'**
1	0	X	X	0	1



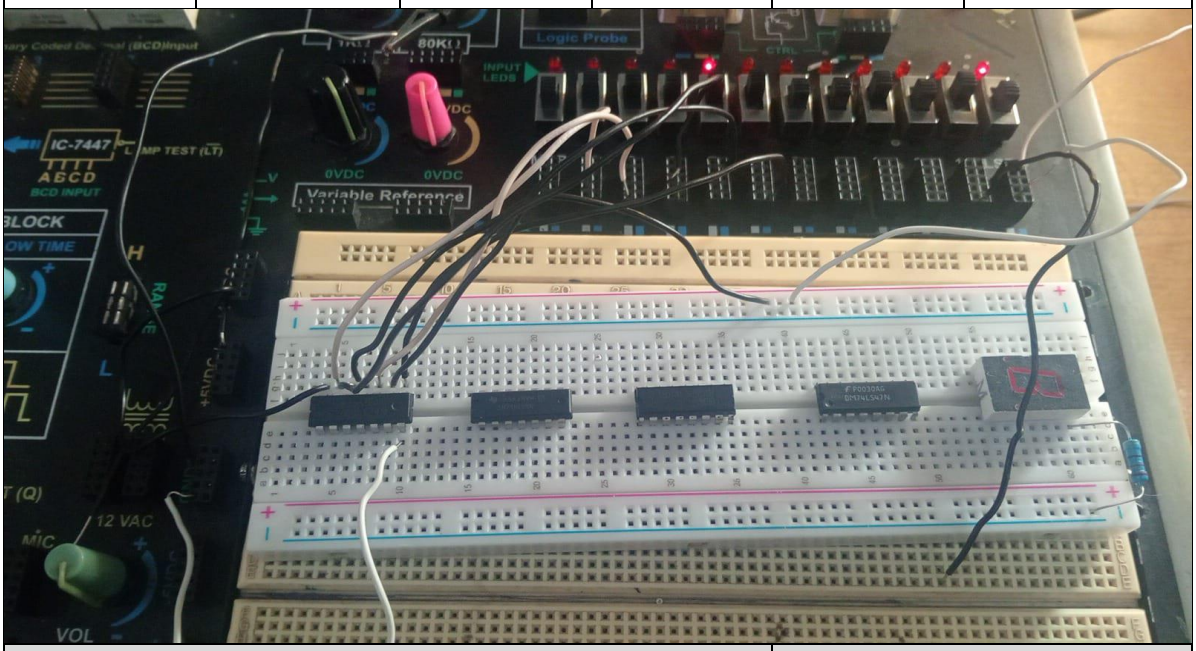
Inputs				Outputs	
PR	**CLR**	**CLK**	**D**	**Q**	**Q'**
0	0	X	X	X	X



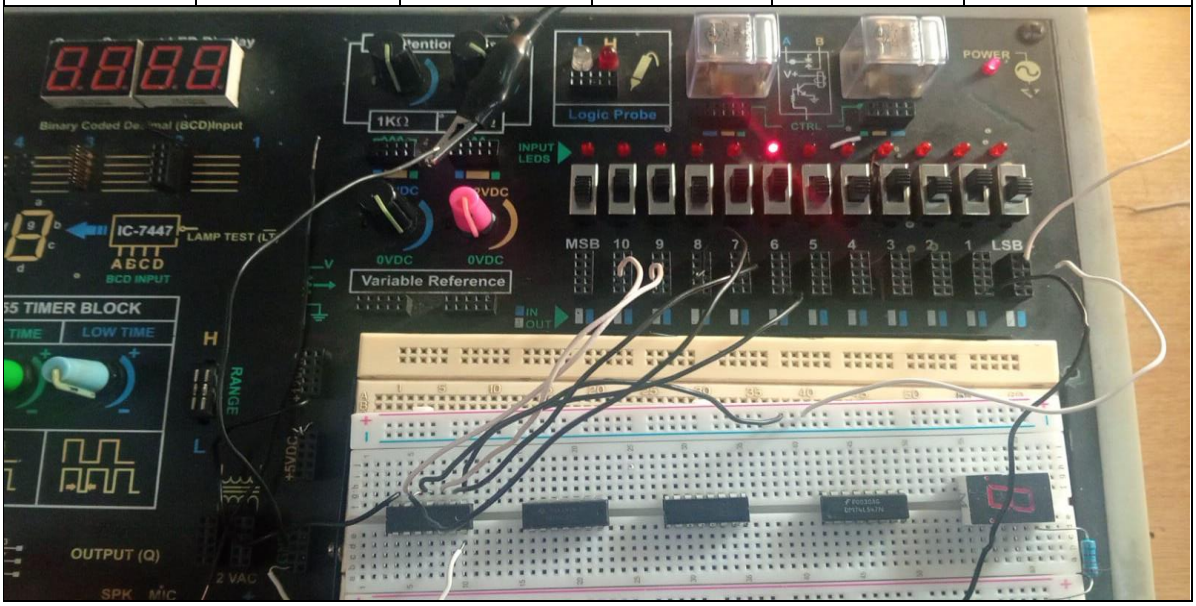
Inputs				Outputs	
PR	**CLR**	**CLK**	**D**	**Q**	**Q'**
1	1	î	X	1	0



Inputs				Outputs	
PR	**CLR**	**CLK**	**D**	**Q**	**Q'**
1	1	î	X	0	1



Inputs				Outputs	
PR	**CLR**	**CLK**	**D**	**Q**	**Q'**
1	1	0	X	Qo	Qo'

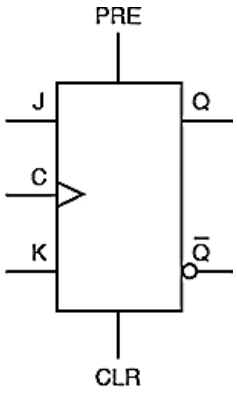


Task 2:

Use JK flip flop from IC 74xx76 and complete its truth table.

Steps:

1. Use 555 timer or CLK on your digital trainer kit as clock source.
2. Observe above waveform on digital oscilloscope.
3. Apply different inputs on the flip flop and verify its results.

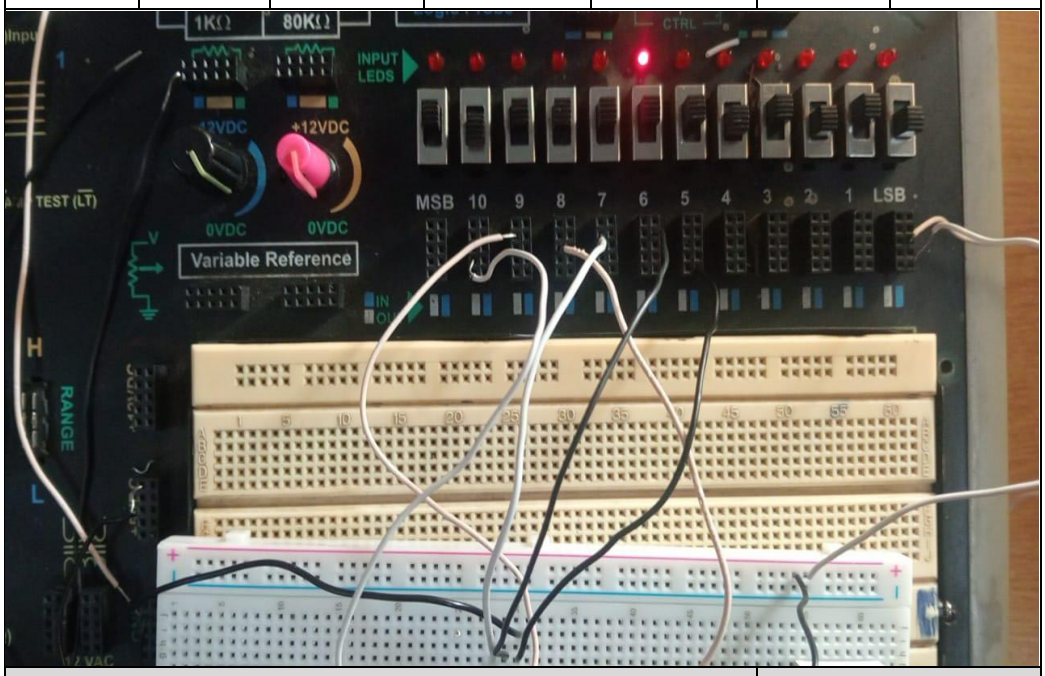


4. Observe outputs on oscilloscope and LED's.

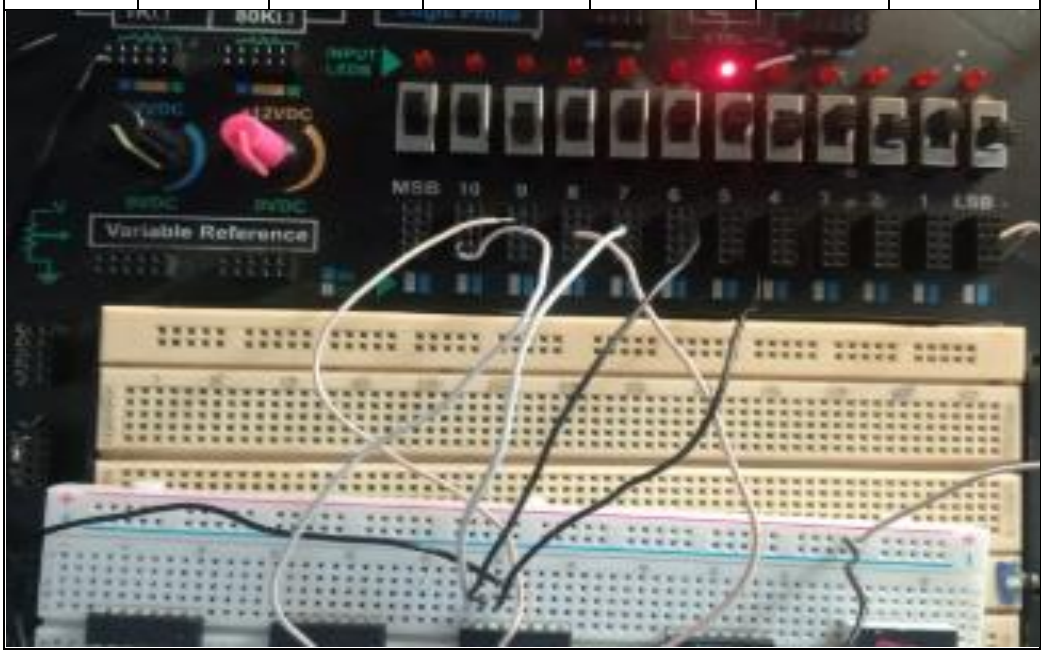
Truth Table:

Inputs					**Outputs**	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
0	1	X	X	X	1	0
1	0	X	X	X	0	1
0	0	X	X	X	1	1
1	1	↑	0	0	Qo	Qo'
1	1	↑	1	0	1	0
1	1	↑	0	1	0	1
1	1	↑	1	1	**Toggle**	

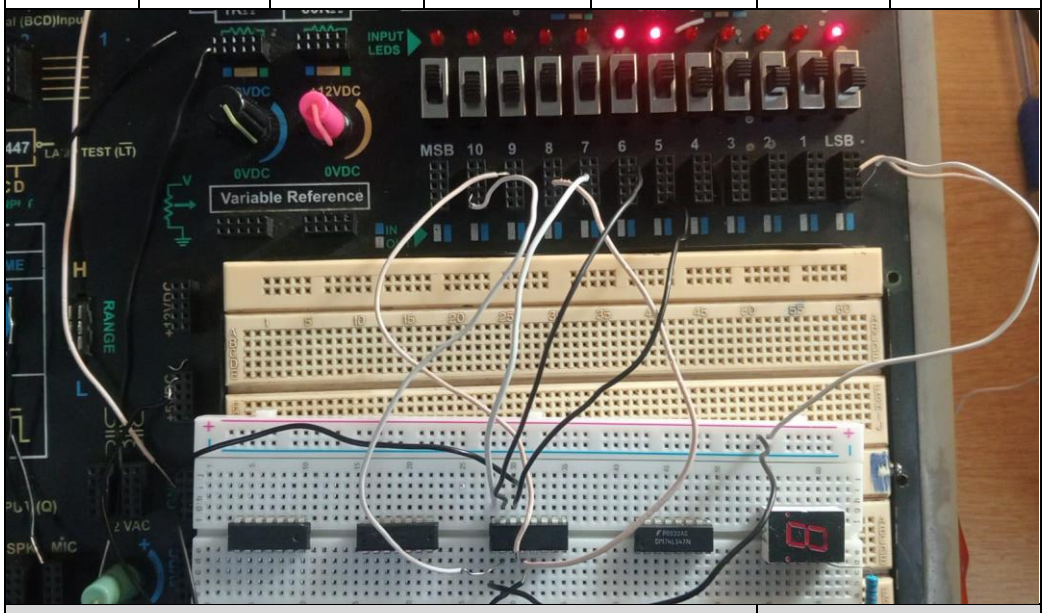
Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
0	1	X	X	X	1	0



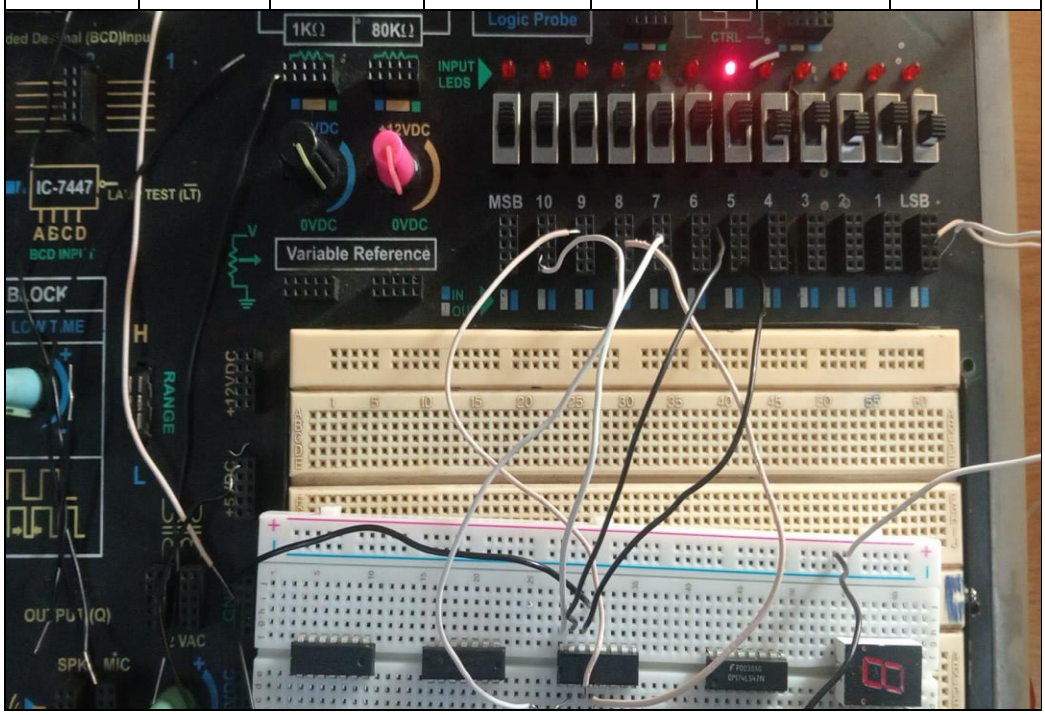
Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
1	0	X	X	X	0	1



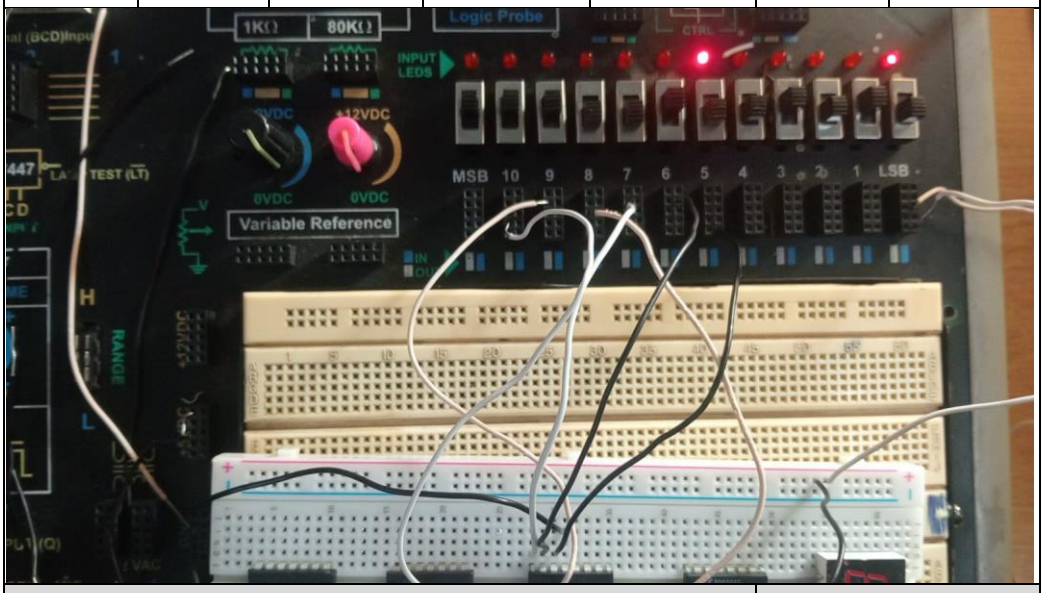
Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
0	0	X	X	X	1	1



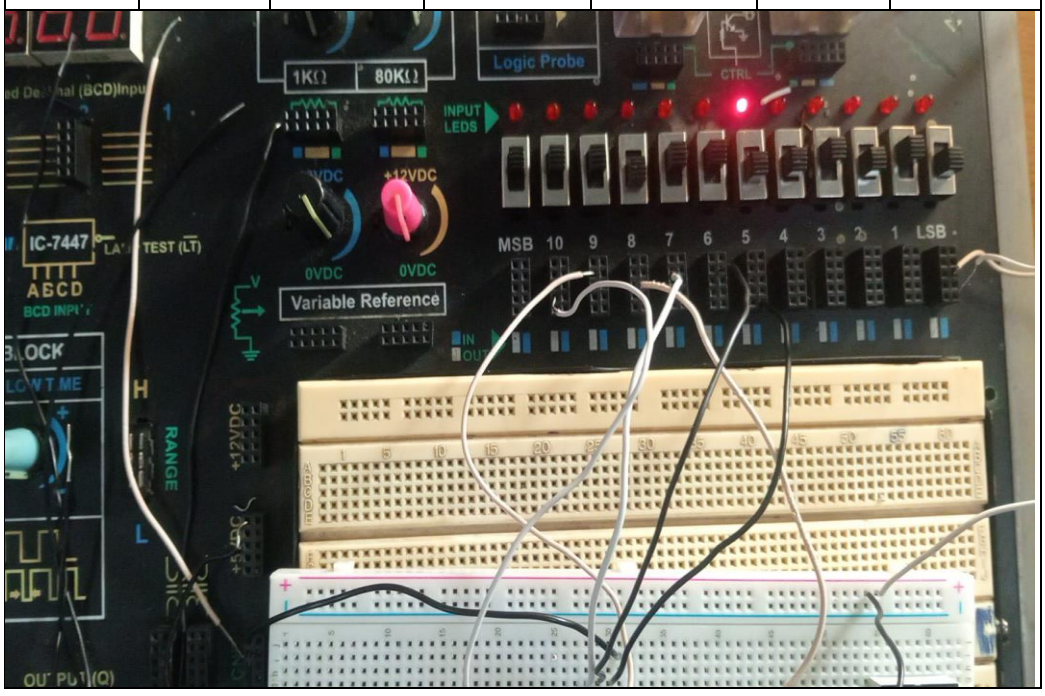
Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
1	1	↑	0	0	Qo	Qo'



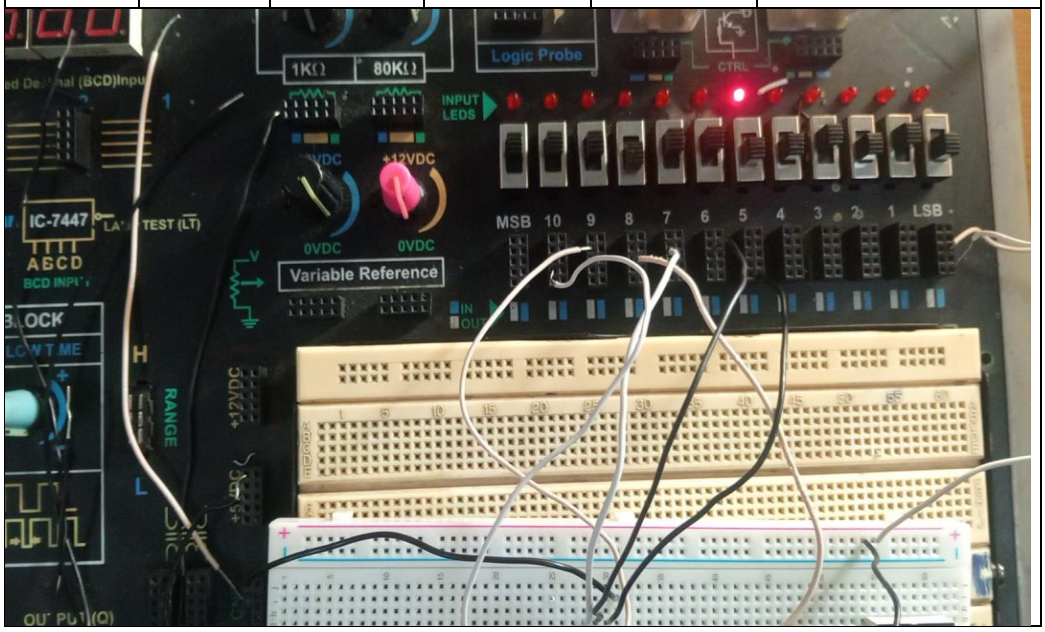
Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
1	1	↑	1	0	1	0



Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
0	1	↑	0	1	0	1



PR	CLR	CLK	J	K	Q	Q’
1	1	↑	0	1	Toggling	



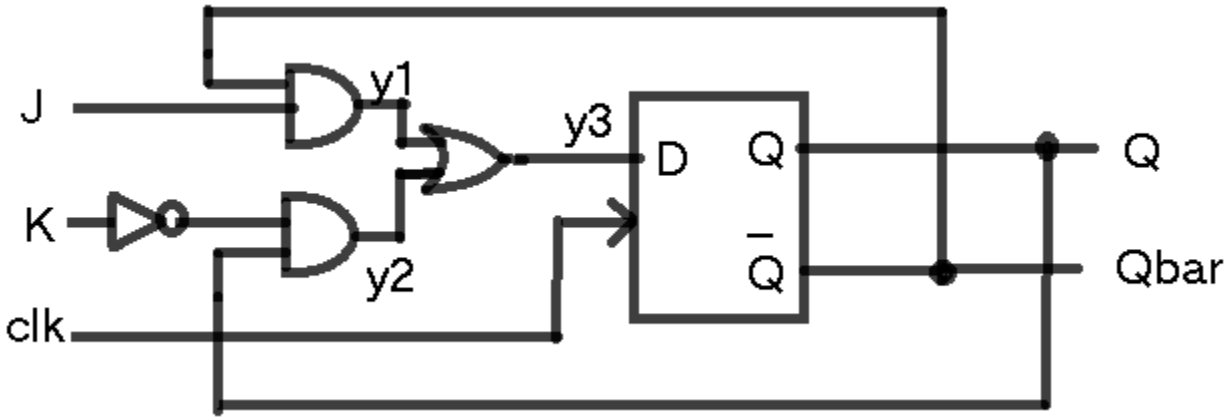
Task 3:

Implement D flip flop using JK flip flop.

Steps:

1. Draw block diagram for D flip flop using JK flip flop.
2. Verify their truth tables and write them.

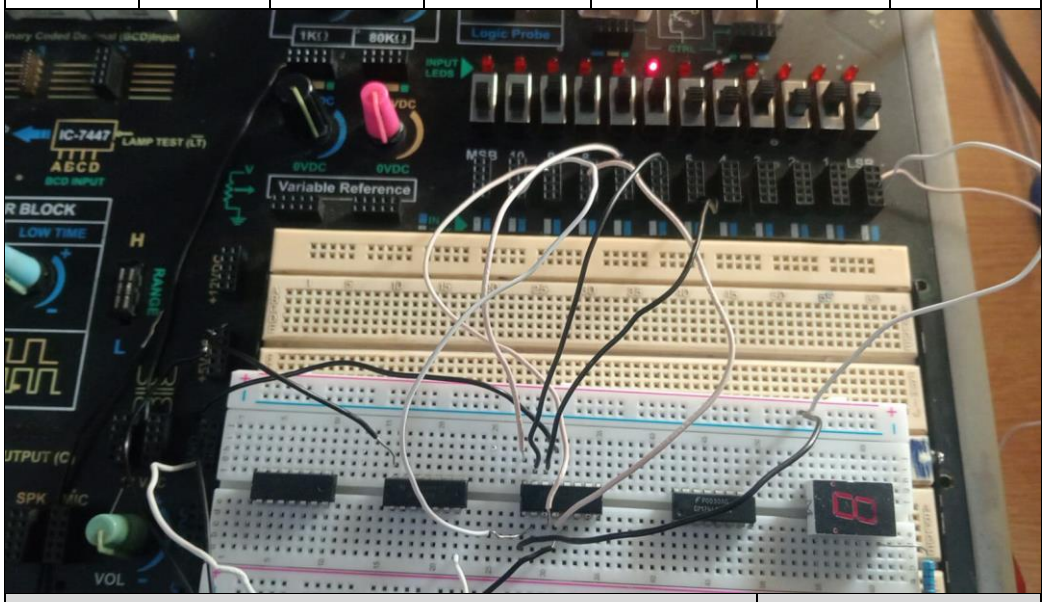
Block Diagram:



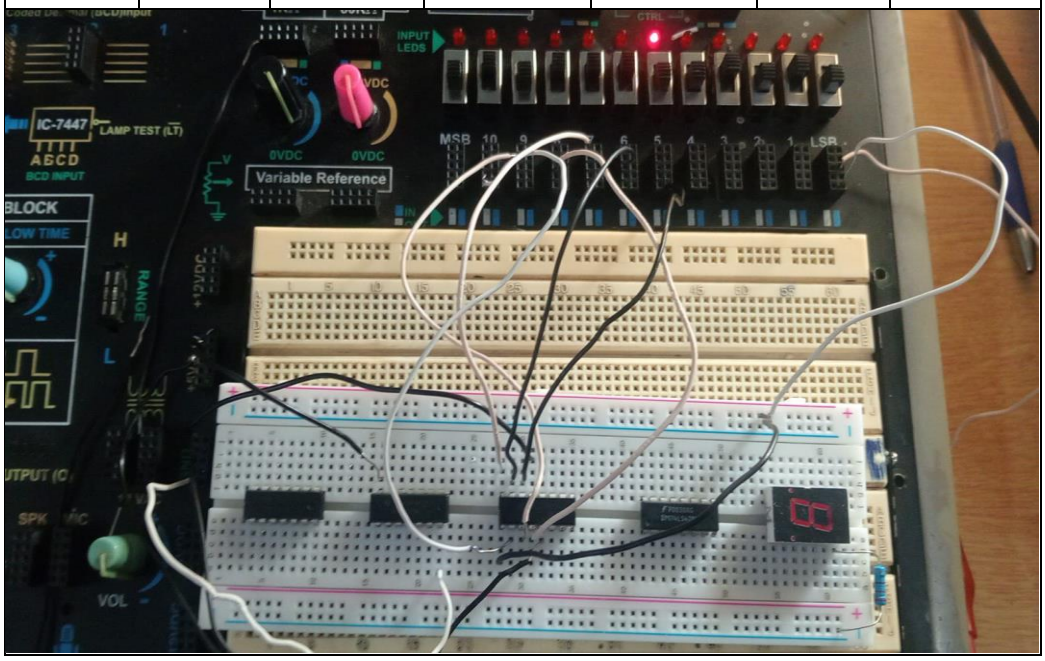
Truth Table:

Inputs					**Outputs**	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
0	1	X	X	X	1	0
1	0	X	X	X	0	1
0	0	X	X	X	1	1
1	1	↑	0	0	Qo	Qo'
1	1	↑	1	0	1	0
1	1	↑	0	1	0	1
1	1	↑	1	1	**Toggle**	

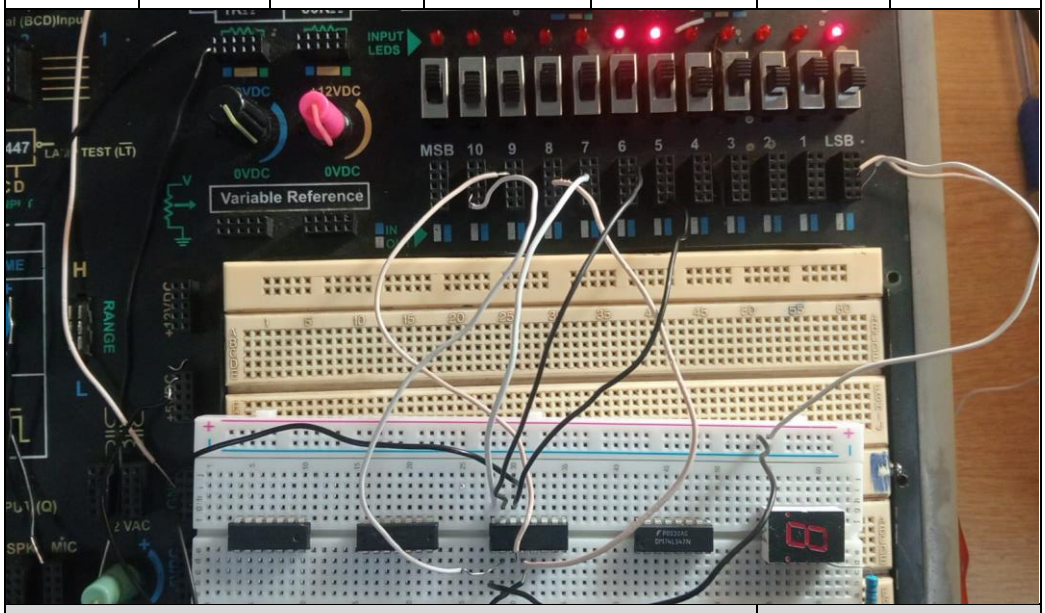
Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
0	1	X	X	X	1	0



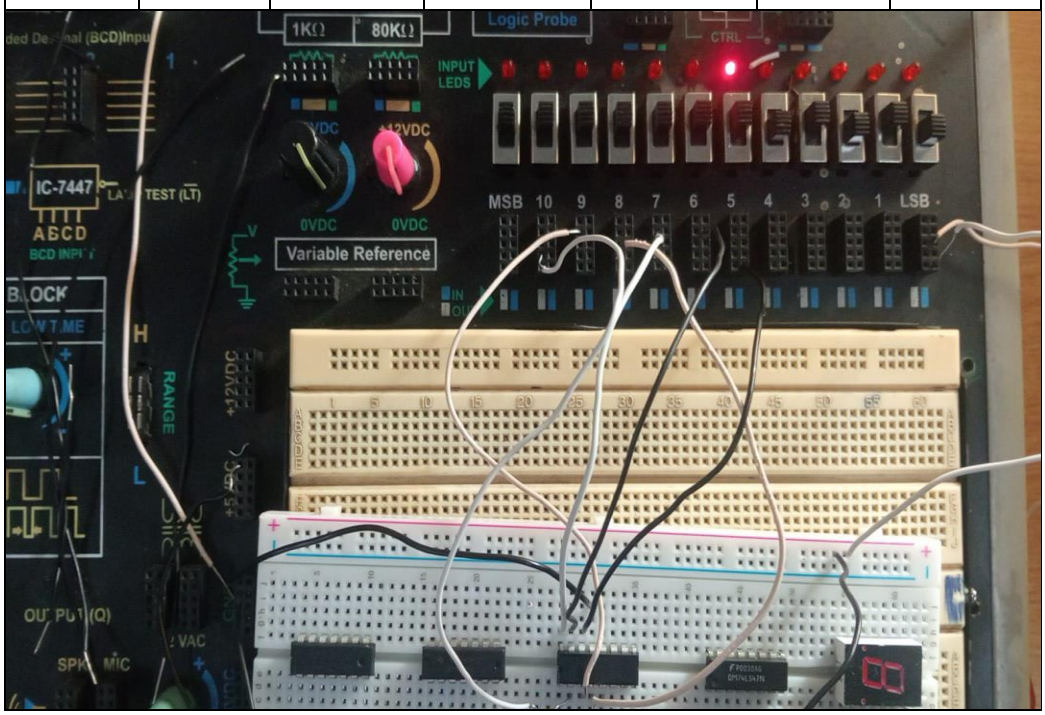
Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
1	0	X	X	X	0	1



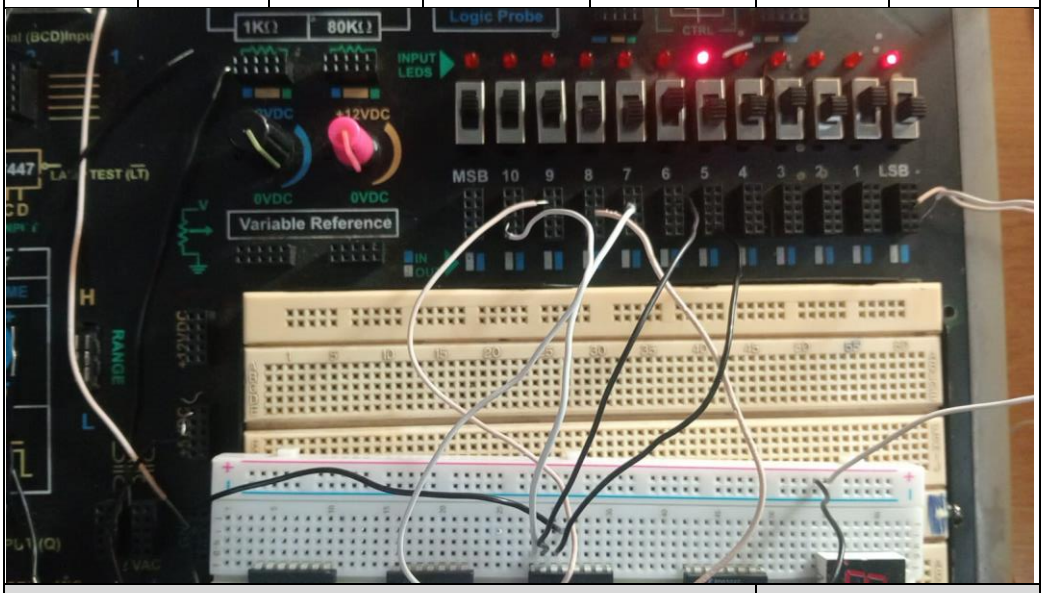
Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
0	0	X	X	X	1	1



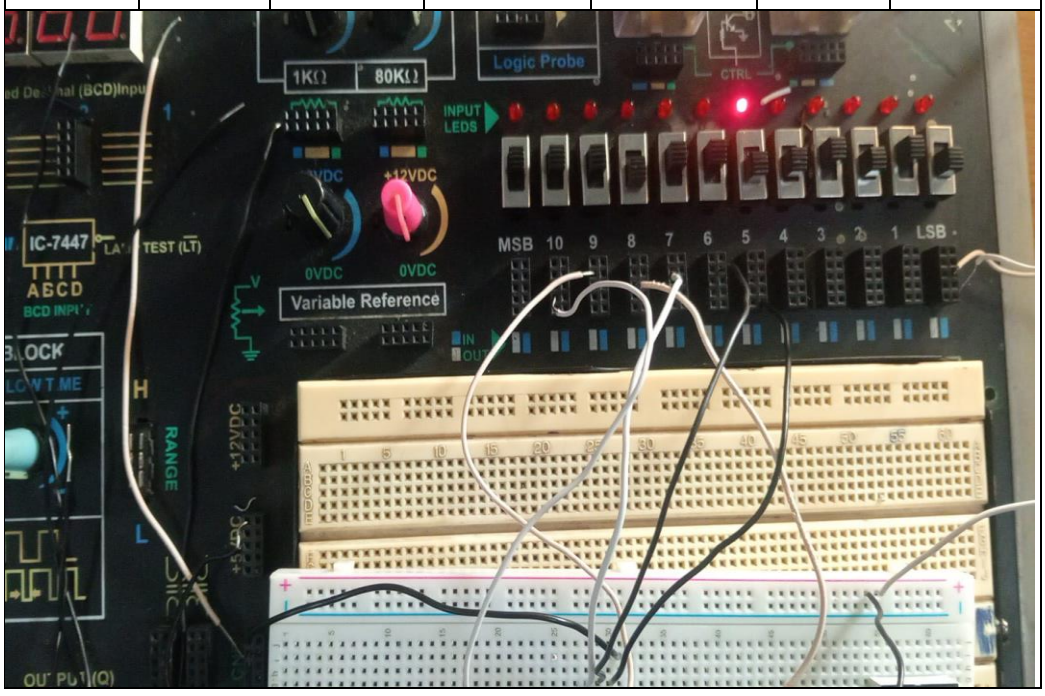
Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
1	1	↑	0	0	Qo	Qo'



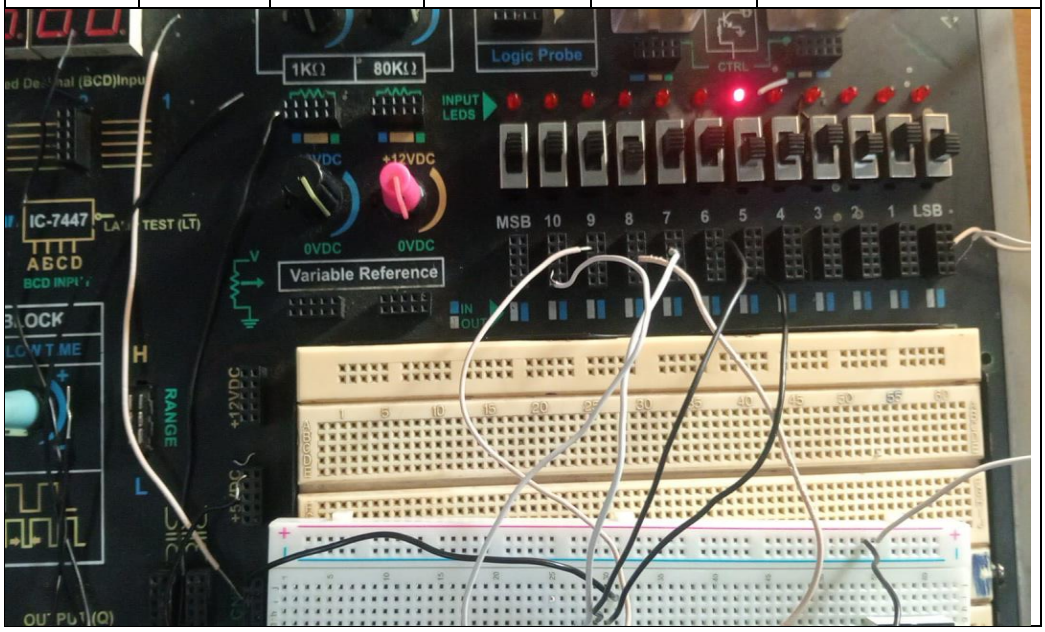
Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
1	1	↑	1	0	1	0



Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
0	1	↑	0	1	0	1



PR	CLR	CLK	J	K	Q	Q’
1	1	↑	0	1	Toggling	



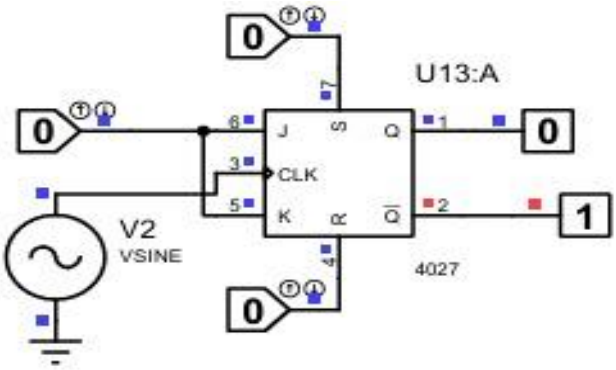
Task 4:

Implement Toggle flip flop using JK flip flop.

Steps:

1. Draw block diagram for Toggle flip flop using JK flip flop.
2. Verify their truth tables and write them.

Block Diagram:

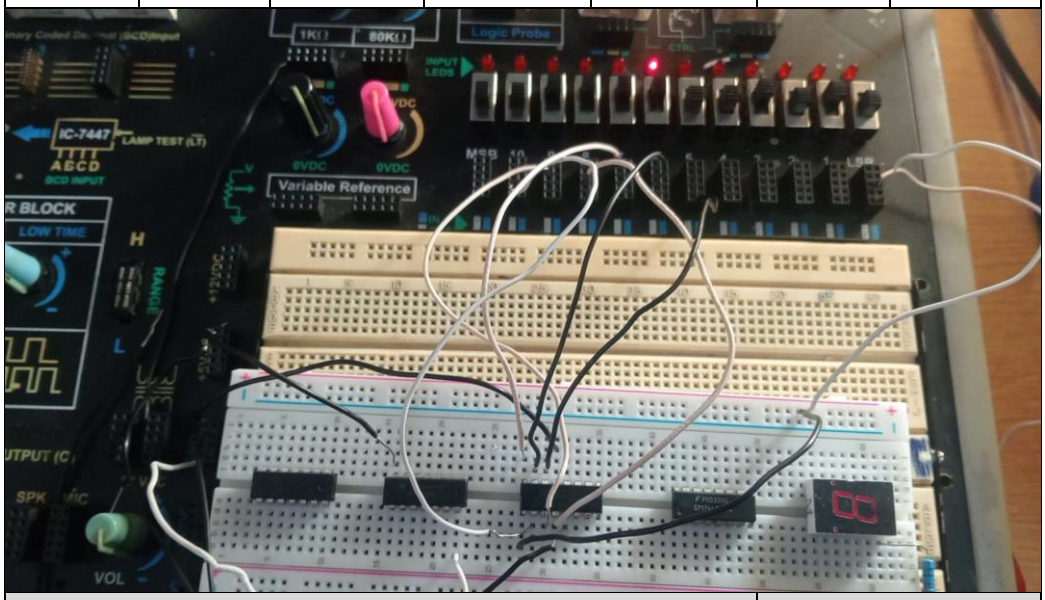


Truth Table:

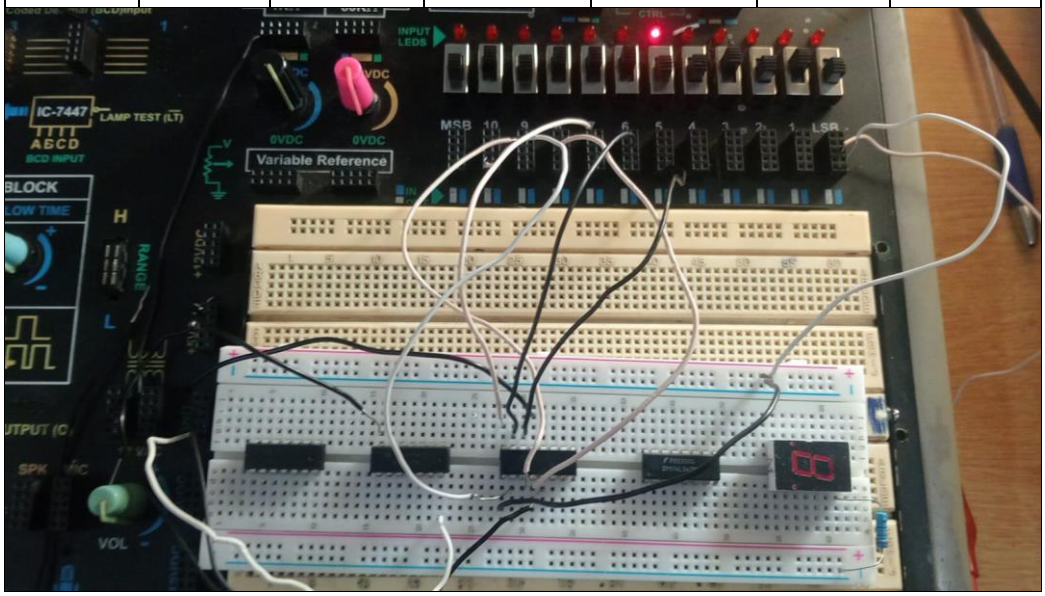
Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
0	1	X	X	X	1	0
1	0	X	X	X	0	1
0	0	X	X	X	1	0
1	1	↑	1	1	Toogle	Toogle
1	1	↑	0	0	Qo	Qo'
1	1	↓	X	X	Qo	Qo'

Simulation Activity:

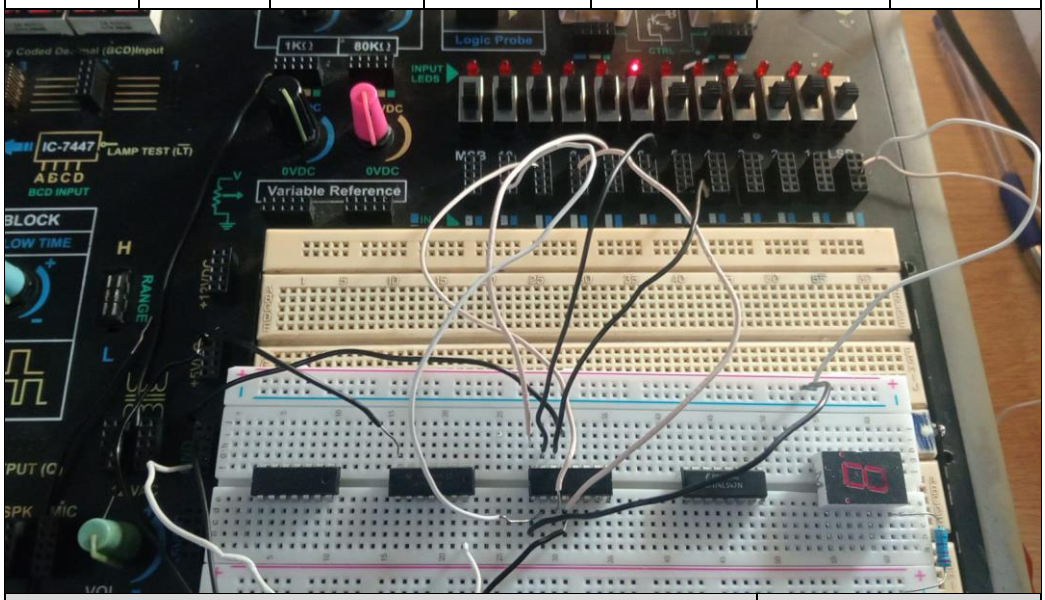
Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
0	1	X	X	X	1	0



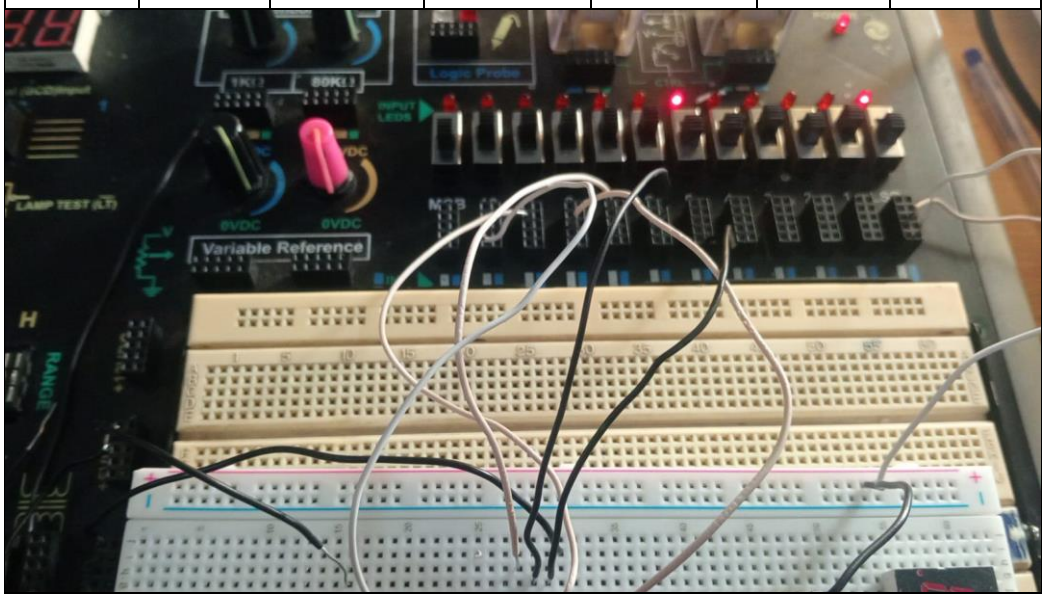
Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
1	0	X	X	X	0	1



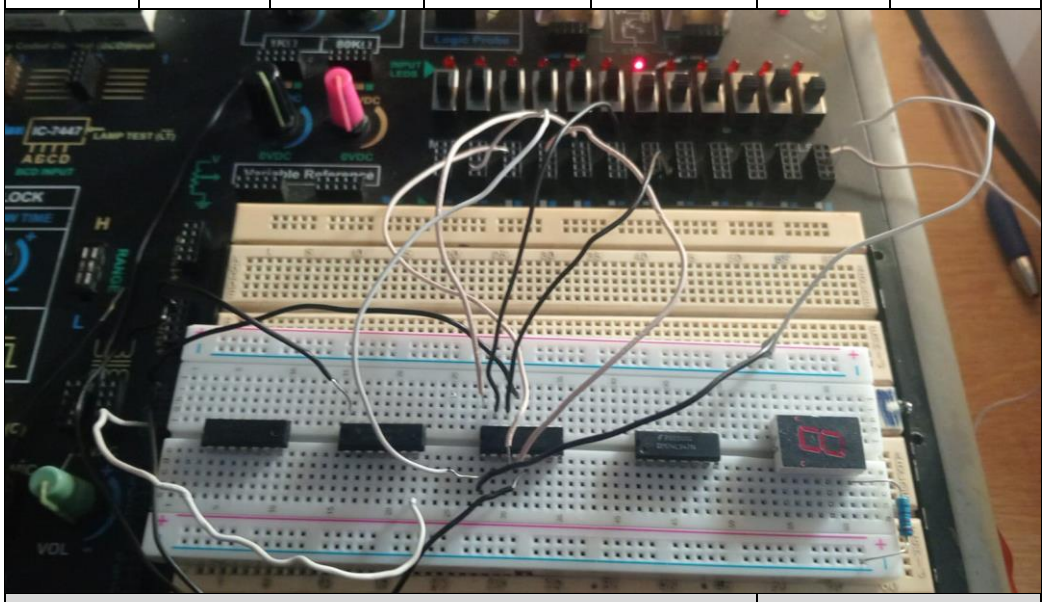
Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
0	0	X	X	X	1	0



Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
1	1	↑	1	1	Toggling	Toggling



Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
1	1	↑	0	0	Qo	Qo'



Inputs					Outputs	
PR	**CLR**	**CLK**	**J**	**K**	**Q**	**Q'**
0	1	↑	X	X	Qo	Qo'

Conclusion:

After performing the experiments on D, JK, and T (toggle) flip-flops in the lab, I understood how each one stores and controls binary data using clock signals. The D flip-flop was the simplest because the output always followed the input D on every clock pulse, making it useful for data storage and registers. The JK flip-flop showed more flexible behavior, as it could set, reset, and toggle depending on the combination of J and K inputs, which helped me observe how multi-purpose control circuits work. When the JK flip-flop was wired with both J and K set to 1, it behaved like a T flip-flop and toggled its output on every clock edge, making it perfect for counters and frequency division. Overall, performing these flip-flops practically helped me clearly understand their operation, switching behavior, and real-time applications in digital systems.